

EC3400 Third Year Economics Dissertation

Volatility Management as Risk Avoidance: Evidence from Cross-Asset Time-Series Momentum

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Abstract

This dissertation examines whether portfolio-level volatility management improves the performance of time-series momentum across a diversified sample of 19 futures contracts from 1990 to 2025. It compares a standard time-series momentum strategy with a volatility-managed alternative based on lagged realised variance. Time-series momentum is still more effective than a passive benchmark in gross terms, but the strategy does not generate statistically significant alpha, and its risk-adjusted performance stays similar to the passive strategy. Even though volatility management appears to initially reduce the high volatility and large drawdowns associated with time-series momentum, the gains reflect a lower average risk exposure rather than improved efficiency. When returns are made equal between the two strategies, the improvements from the volatility managed strategy disappear. Once transaction costs are added, the strategy performs even worse under relatively modest cost assumptions. Therefore, it implies that variance timing in this scenario acts more as a form of risk avoidance than strategy improvement.

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1. Introduction

In its semi-strong form, the efficient market hypothesis (EMH) implies that public information should be rapidly incorporated into asset prices, leaving little scope for abnormal returns that are based on past price movements (Fama, 1970). Yet one of the most influential empirical regularities in financial markets is momentum: the tendency for assets with strong recent performance to continue to outperform over intermediate horizons, whilst recent losers continue to underperform. Jegadeesh and Titman (1993) show that zero-cost equity portfolios formed by buying recent winners and selling recent losers generate significant profits over horizons of roughly three to twelve months.

Whilst this evidence primarily concerns cross-sectional momentum, a related and practically important strategy is time-series momentum. Rather than ranking assets relative to one another, time-series momentum uses an asset's own past return, taking a long position following positive past performance and a short position following negative past performance. This makes it especially relevant in broad futures markets, where it forms a substantial portion of the trend-following literature.

The key issue, however, is not simply whether momentum exists, but whether it remains attractive once its risk profile is examined more carefully. Strategies based on momentum signals can be inherently risky, as they are exposed to periods of severe underperformance, particularly following abrupt market reversals. Daniel and Moskowitz (2016) show that momentum crashes are partly predictable and tend to occur in periods of elevated volatility associated with rapid recoveries. Similarly, Barroso and Santa-Clara (2015) argue that momentum risk varies substantially through time and that exposure adjustments based on this variation can reduce crash sensitivity and improve risk-adjusted returns. Together, these contributions shift attention away from whether momentum is profitable on average and towards whether its returns are sufficiently stable and investable in practice, thereby introducing the case for volatility management.

The intuition behind this is relatively straightforward. If volatility is persistent over time, strategies that reduce their risk exposure when markets are turbulent and increase exposure when conditions are more stable may achieve a more favourable risk-adjusted return than strategies that maintain constant exposure during differing market environments. This raises the central question of this dissertation: does volatility management genuinely improve time-series momentum, or does it simply produce a smoother, lower-exposure version of the same underlying strategy?

It is important to clarify what exactly is being tested. The volatility managed approach that is used in this dissertation follows Moreira and Muir's (2017) specification, which applies variance scaling to an equally weighted momentum strategy that uses the sign of the signal to determine trading positions, rather than acting as an overlay on top of the base TSMOM portfolio, which scales volatility across its assets universe. The comparison should therefore be interpreted as an assessment of whether this particular and widely regarded form of

variance timing improves on a standard and diversified momentum strategy, rather than as a pure evaluation of the overlay's sole effect.

This dissertation sits at the intersection of these literatures by examining whether time-series momentum remains attractive as a trading strategy in a contemporary setting, and whether volatility management improves its stability and performance. Revisiting these issues is particularly important in the post-global financial crisis and post-pandemic environment, where macro-financial conditions have changed significantly. In such a setting, it is not obvious that either momentum strategies or volatility scaling will continue to deliver the performance documented in earlier samples.

The argument in favour of re-examining these questions rather than relying on the results from the literature comes from the recent economic environment that previous samples could not have captured. In recent years, trend-following performance has reportedly weakened in the years since the global financial crisis compared to the decades before it. Hurst, Ooi and Pedersen (2017) show that trend-following strategies have been very profitable across a multitude of historical market conditions, but even their long-run sample cannot attest to how the strategy has performed under the market environment that has prevailed since 2009. This ultimately raises a reasonable question surrounding the strong results that are reported by Moskowitz, Ooi and Pedersen (2012) and that is whether they continue to hold in a contemporary sample.

A second reason relates more to how the strategy itself works. A large part of how returns were historically generated in trend-following comes from the long positions on government bonds, which massively benefitted from the decline in global bond yields over multiple decades. Once rates in major economies reached their effective minimum and central banks resorted to asset purchases on a large scale, the scope for sustained bond trends that drove earlier performance was naturally reduced. This does not ultimately conclude that trend-following must have weakened, but it does mean that one of its most reliable drivers may have been at a diminished strength for this particular sample.

Thirdly, the post-pandemic inflationary shock as well as the shock to interest rates represents a genuine structural shift rather than a continuation of the previous regime. The rapid shift from near-zero rates to a relatively aggressive tightening cycle, combined with the first sustained inflation episode in several decades, is unlike anything in the Moskowitz, Ooi and Pedersen (2012) sample. Whether the behaviour of time-series momentum remains the same in this regime as it was in the previous one is not something that the prior literature can answer, and it is one of the most intriguing questions a re-examination can address.

This matters not only for whether TSMOM remains an attractive trading strategy, but also for whether volatility management can improve it. If the underlying strategy is less reliable, then maybe an overlay that was designed to stabilise it could behave differently as well. McLean and Pontiff (2016) show that many return predictor strategies that are alike to time-series momentum weaken out of sample. This suggests that modern re-evaluations of different strategies are not redundant even when the initial evidence is strong. This dissertation addresses these concerns directly by examining whether time-series momentum continues to

deliver meaningful performance in a modern setting, and whether volatility management can genuinely improve the strategy.

The remainder of the dissertation is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the data and variables. Section 4 outlines the empirical methodology. Section 5 presents the main results. Section 6 forms the discussion. Section 7 concludes, Section 8 contains the references, and Section 9 the appendix.

2. Literature Review

The empirical literature most directly related to this dissertation begins with Moskowitz, Ooi and Pedersen (2012), whose specification provides the central benchmark for time-series momentum. Using a sample of 58 liquid instruments that comprise equities, government bonds, commodities, and currency futures, they show that an asset's own past return predicts the direction of its future return, with continuation strongest at lookback horizons of one to twelve months and partial reversal at longer horizons. They show that, across asset classes, time-series momentum delivers an annualised Sharpe ratio above 1.0, which is roughly 2.5 times the standard equity market portfolio. This positions time-series momentum as a systematic trend-following strategy rather than purely a stock market anomaly. Within commodity futures specifically, Miffre and Rallis (2007) show that momentum strategies generate significant profits across 31 commodity contracts, with the returns remaining largely uncorrelated to traditional asset classes. Their findings give commodity specific evidence for the return continuation patterns present across asset classes. The view that time-series momentum acts as a component of broader cross-asset systematic premia is reinforced by Koijen, Moskowitz, Pedersen and Vrugt (2018). They show that carry strategies produce large returns across the same set of asset classes used in the original time-series momentum literature, and similarly has modest correlation to the risk factors implemented. This places trend-following alongside, rather than in competition with other systematic futures-based strategies.

Subsequent research broadly reinforces this conclusion. Hurst, Ooi and Pedersen (2017) extend the evidence over a substantially longer historical period, examining 67 liquid markets from 1880 to 2016. They show that trend-following remains profitable across a wide range of macroeconomic conditions, whilst maintaining relatively low correlation with traditional asset classes. An earlier paper from Fung and Hsieh (2001) aligns well with this characterisation. They show that the returns of hedge funds that regularly employ trend-following strategies can be closely replicated by portfolios of lookback straddles, formalising the view that the payoffs surrounding trend-following are similar to options and that linear regression models are at risk of understating their economic value. This strengthens the view of time-series momentum as a meaningful diversifier rather than a period-specific anomaly. Baltas and Kosowski (2013) similarly show that trend-following has an important role in explaining CTA index returns, whilst Hamill, Rattray and van Hemert (2016) further emphasises its tendency to perform well in adverse market conditions. Clare, Seaton, Smith and Thomas (2016) similarly find that a combination of trend-following and risk-parity allocation across asset classes improves risk-adjusted performance against relative benchmarks, reinforcing the argument in favour of trend-following as a portfolio level implementation rather than purely as a single asset signal. Taken together, this literature suggests that time-series momentum is historically broad-based, scalable, and attractive from an institutional perspective.

Yet the strength of this evidence also creates an important problem. Much of the literature focuses strongly on the long-run profitability of time-series momentum, but is much less concerned with whether the strategy remains attractive once the distribution of returns through time is examined more carefully. A strategy may exhibit strong average returns whilst still being difficult to justify as an investment approach if those returns are accompanied by severe drawdowns or unstable payoffs. That time-series momentum works on average is therefore not, by itself, evidence that it is robust or straightforward to implement in practice.

The strength of the original time-series momentum evidence has been qualified by several subsequent contributions that question either its sources or its robustness. A more skeptical view of these findings is provided by Kim, Tse and Wald (2016), who argue that the apparent profitability of time-series momentum is to a large extent down to the weights used for volatility-scaling when constructing the strategy rather than to an actual return continuation. They show that once positions are held without scaling by inverse volatility, the marked outperformance over a simple passive benchmark is substantially reduced. This is directly relevant to this present setting, where volatility scaling is used on both the cross-asset position level and through the variance-management overlay. Huang, Li, Wang and Zhou (2020) offer a complementary critique. They document that the original time-series momentum results weaken substantially when alternative test specifications are used in place of the pooled-panel regressions from Moskowitz, Ooi and Pedersen (2012).

Daniel and Moskowitz (2016) address this issue directly by focusing on the state-dependent risk of momentum strategies. They show that momentum crashes are concentrated in market rebounds following periods of stress, suggesting that momentum may be profitable in expectation whilst remaining fragile to rapid reversals.

Barroso and Santa-Clara (2015) provide the most direct precedent for the approach taken in this dissertation. They argue that a large part of the crash risk associated with momentum strategies can be managed by information that is present in a strategy's own recent return history. Their specification does not forecast expected returns or broader market conditions, they instead use the fact that the realised variance of the momentum strategy is highly persistent over time but only correlates weakly with its own expected return, so that exposure can be lowered during more volatile periods and heightened when the volatility is low. Therefore the behavior of the strategy's own recent return series is the basis of its own risk management. Specifically for the momentum of U.S. based equities, they show that a version that keeps volatility constant on average halves the occurrence of severe drawdowns and produces a Sharpe ratio nearly twice the value of the unmanaged benchmark. What makes this especially relevant to this dissertation is that the scaling applies directly to the momentum strategy itself instead of applying it to a wider portfolio. This enables a better test of volatility management than earlier studies on volatility-timing, because the overlay's effect can be interpreted in a more direct sense. It also ties the paper closer to this dissertation, which also questions whether adding a layer of volatility management can genuinely improve momentum. The natural continuation of this is whether the same principles carry over to a multi asset evaluation of time-series momentum, where the underlying strategy already uses

volatility scaling across assets when it takes positions and may as a result leave less unmanaged variance for a portfolio-level addition to improve upon.

A related strand of literature asks whether this fragility can be alleviated by adjusting risk exposure in response to changes in volatility. The basis for this idea comes from the literature on time-varying conditional variance. The ARCH model by Engle (1982) and Bollerslev's (1986) GARCH extension formalise the view that volatility is persistent over time and is therefore at least partly predictable. At the base of the portfolio, Fleming, Kirby and Ostdiek (2001) show that volatility timing can still add value even when expected returns are themselves difficult to predict. This motivates adjusting the exposure to risk dynamically, even at times where the return signal itself does not change.

Moreira and Muir (2017) use the same structure but in a wider context by applying volatility timing to numerous equity factor portfolios as opposed to using solely one strategy. Their results show that scaling portfolios inversely to recent realised variance can lead to higher Sharpe ratios and statistically significant alphas. Their argument is that factor volatility tends to vary more than expected returns, which implies that keeping exposure constant through time could be inefficient. The paper therefore relates to Barroso and Santa-Clara (2015) but it acts as more of an extension of their findings, which is why it has become so important in the literature of managing variance. However, it is important to differentiate between the portfolio-level volatility management used by Moreira and Muir (2017) from the cross-asset risk normalisation built into momentum strategies like time-series momentum by Moskowitz, Ooi and Pedersen (2012). Volatility management changes the overall exposure of the strategy according to recent volatility of the entire portfolio, whereas time-series momentum scales each asset by its own ex-ante volatility so that a single market doesn't dominate the portfolio. These two methods are evidently not the same, and whether a result established for the first carries over to a setting where the latter is present is something that needs to be tested

Recent studies have argued in favour of using volatility management. Wang and Yan (2021) show that scaling a strategy's exposure by downside volatility, rather than total volatility leads to stronger improvements in factor-portfolio performance. A large part of this improvement is down to return timing rather than solely volatility timing. They also factor in how implementable the results are in a real-time trading setting, which is important because it reinforces the direct practicality of the strategy to an investor. Schwarz (2025) gives a more cautious outlook by showing that the supposed gains reported in earlier literature suffer once volatility managed equity factors are examined internationally, and that many of the gains weaken once transaction costs are accounted for. Cederburg, O'Doherty, Wang and Yan (2020) offer a more direct challenge to the volatility management literature. They show that the benefits from Moreira and Muir (2017) mostly fail to replicate out of sample across a wide range of equity-based factors. Their evidence moves the literature to more of a skeptical view of variance timing as an effective way of improving the performance of a portfolio.

The papers however do not overturn the results from Moreira and Muir (2017) and Barroso and Santa-Clara (2015), but they do suggest that how beneficial variance timing is dependent on the specific setting and how the strategy is implemented. This is particularly useful for this

dissertation as it asks the same questions in an alternative setting and uses transaction costs as an important robustness check.

Even where the case for volatility management appears strong, its empirical interpretation is not straightforward. A reduction in volatility or drawdown may simply reflect lower average risk exposure rather than a genuine improvement in strategy efficiency. For that reason, return comparability is essential. If a volatility-managed strategy earns lower average returns because it systematically lowers its exposure to risk, then a raw comparison of Sharpe ratios or drawdowns may overstate its benefits. Volatility management tends to require frequent rebalancing, which can materially increase trading costs. These issues are especially important in the present context because they determine whether volatility management genuinely improves the strategy's risk-adjusted performance.

A further issue concerns how momentum profits should be interpreted more broadly. Barberis, Shleifer and Vishny (1998), Daniel, Hirshleifer and Subrahmanyam (1998), and Hong and Stein (1999) each propose behavioural mechanisms that are centred around underreaction, overconfidence, or gradual information diffusion, thereby generating return continuation in the medium term and reversal over longer horizons. A competing view is that momentum reflects compensation for certain risks rather than pure mispricing. Another strand of the literature questions whether profits remain robust once practical objections are taken seriously. Asness, Frazzini, Israel and Moskowitz (2014) also address several common critiques of momentum strategies, this includes concerns surrounding implementation costs, weaker effectiveness after publication, and tax efficiency, and they argue that the return premium remains robust to many of the objections raised in the literature. However, their evidence concentrates on equities, leaving the door open to whether this extends to multi-asset futures settings.

The literature also provides reasons for caution when revisiting established return predictors. McLean and Pontiff (2016) show that many published anomalies weaken out of sample and after publication, whilst Harvey, Liu and Zhu (2016) argue that the large number of tested factors in empirical finance makes conventional significance thresholds too permissive. Although neither paper is specific to time-series momentum, both are relevant here. They imply that re-evaluations of prominent strategies are not redundant, but form part of the empirical burden of proof. A related issue is raised by Novy-Marx (2012), who argues that the standard momentum signal is driven by medium horizon returns, so around 7-12 months, rather than by recent past performance. Whilst his analysis centres on cross-sectional equity momentum, the additional implication that the lookback period chosen meaningfully affects the profitability of momentum is especially relevant to this paper's specification, which uses a lookback period of 12 months.

The literature ultimately points to three rather clear findings but also leaves one important question unresolved. There is strong evidence to suggest that time-series momentum is a profitable investment strategy across various asset classes over the long run, this is confirmed by both Moskowitz, Ooi and Pedersen (2012) and Hurst, Ooi and Pedersen (2017). It is also evident that momentum is exposed to crash risk, and that part of this risk can be anticipated

from a strategy's own past behaviour (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015). Additionally, scaling exposure inversely to recent realised variance can improve equity factor portfolios (Moreira and Muir, 2017; Wang and Yan, 2021; Schwarz, 2025). What is somewhat less evident is whether volatility management can be applied to cross-asset time-series momentum and deliver similar performance improvements. The present context is particularly unique as the base strategy already applies volatility scaling at the position level. Therefore, the key question is if variance timing at the portfolio level can bring added value to the strategy once differences in average exposure and transaction costs are considered.

3. Data and variables

Bloomberg settlement prices on generic continuous futures series are used when constructing the data to test the effectiveness of time-series momentum across a wide range of assets. Since individual contracts are subject to expiry, analysing return continuation requires data that stretches across successive maturities. Settlement prices provide consistent end-of-day price valuations and avoid intraday trading noise; they are also the standard convention when researching futures-based strategies.

The data consist of futures prices for 19 different assets, including 8 commodities, 4 equity indices, 3 government bonds, and 4 currency pairs, from January 1990 to December 2025. The panel is unbalanced, with contracts entering the sample when reliable settlement price histories begin, reflecting differences in market histories rather than discretionary sample selection.

This cross-asset design provides a demanding setting for time-series momentum. Equity indices, bonds, commodities, and exchange rates are driven by different macro-financial mechanisms and may exhibit varying degrees of trend persistence. Evidence of return continuation across these heterogeneous markets therefore constitutes stronger support for time-series momentum than findings confined to a single asset class (Moskowitz, Ooi, and Pedersen, 2012; Asness, Moskowitz, and Pedersen, 2013). This is especially relevant in regards to the commodity component of the sample, where Gorton, Hayashi and Rouwenhorst (2013) show that returns are based on fundamentals that structurally differ from the drivers of equity and fixed-income returns. Using commodities in the sample therefore introduces a return producing process that is not similar to the wider macro-financial environment.

Daily settlement prices are used for volatility estimation, whilst monthly returns evaluate the overall strategy performance. This split is common in the literature, as volatility responds to higher-frequency information, whereas the return predictability hypothesis is stated at the monthly horizon according to Moskowitz, Ooi and Pedersen (2012).

The primary inputs consist of futures returns calculated from monthly settlement prices. The momentum signal is based on the lagged performance of an asset. Contracts with returns that were previously positive are assigned a long position, and contracts with negative past returns are assigned a short position. This differentiates time-series momentum from cross-sectional momentum, as positions depend on absolute past performance rather than their relative performance.

Since volatility differs noticeably across assets, an ex-ante volatility estimate is used to scale positions in order to prevent the most volatile contracts from dominating the portfolio. Following Moskowitz, Ooi, and Pedersen (2012), the estimator is an exponentially weighted average of lagged squared daily returns, annualised and applied with a one-period lag to eliminate look-ahead bias. Volatility scaling serves both as a necessary normalisation across

assets and as a distinct strategy feature, enabling a direct comparison between the time-series momentum and volatility-managed implementations.

Strategy performance is evaluated using the monthly excess returns relative to the 1-month U.S. Treasury bill, which acts as the risk-free rate in the analysis, consistent with Fama and French's (1993) standard protocol for asset pricing. To test whether apparent profitability reflects momentum-related alpha rather than sensitivity to common risk factors, the regressions include a set of baseline controls: the MSCI World equity excess return as the market factor, SMB and HML from the Ken French Data Library, the UMD momentum factor, and broad bond and commodity benchmarks. This follows the same factor structure as Moskowitz, Ooi, and Pedersen (2012) and Asness, Moskowitz, and Pedersen (2013).

Table 1 summarises all the key variables, and Table 2 reports the annualised returns and volatilities for each contract. Two patterns stand out. First, there is substantial variation across contracts. Natural gas is the clearest outlier, with an annualised return of 17.53% and volatility of 54.82%, whilst soybeans return just 0.53% at 8.01% volatility. Second, this variation is largely systematic across asset classes rather than idiosyncratic. Equity index futures such as the S&P 500 record returns of 8.24% with volatility of 15.44%; government bond futures are relatively more stable. Currency futures display the most restrained average returns, with several contracts close to zero across the full sample.

TABLE 1 – SUMMARY OF KEY VARIABLES

Variable	Description
Monthly futures return	Monthly percentage change in futures settlement prices.
Daily futures return	Daily percentage change in futures settlement prices.
Passive benchmark return	Return on the passive benchmark portfolio across available contracts.
TSMOM signal	Long-short trading signal based on the sign of the previous 12-month return, lagged by one month.
Ex ante volatility	EWMA volatility estimate from daily returns, used to scale individual asset positions in TSMOM.
TSMOM return	Return on the baseline cross-asset time-series momentum strategy with volatility-scaled positions.
VM-TSMOM return	Return on the volatility-managed momentum strategy, where the base momentum return is scaled by inverse lagged realised variance.
Realised variance	Monthly realised variance of the base momentum strategy, estimated from squared daily portfolio returns.
Scaling weight	Time-varying portfolio weight applied in the volatility-managed strategy, based on inverse lagged realised variance.
Risk-free rate	One-month U.S. Treasury bill rate used to calculate excess returns.
Sharpe ratio	Mean excess return divided by return volatility.
Maximum drawdown	Largest peak-to-trough decline in cumulative portfolio value over the sample.
Turnover	One-way monthly portfolio turnover, calculated from changes in portfolio weights.
Transaction costs	Costs deducted from returns using turnover-based basis point assumptions.
MKT	Market factor proxied by MSCI World excess returns.
SMB	Fama-French size factor.
HML	Fama-French value factor.
UMD	Cross-sectional momentum factor.
BOND	Broad bond market benchmark return.
GSCI	Broad commodity market benchmark return.

Notes: Main variables used in the analysis

TABLE 2 – FUTURES CONTRACTS IN THE SAMPLE WITH SUMMARY STATISTICS

Contract	Data start date	Annualised return	Annualised volatility
<i>Commodity futures</i>			
COPPER	Feb-90	8.03%	24.66%
CORN	Feb-90	5.44%	27.47%
CRUDE OIL	Feb-90	6.97%	33.44%
GOLD	Feb-90	7.94%	15.24%
NATURAL GAS	May-90	17.53%	54.82%
SILVER	Feb-90	11.62%	28.76%
SOYBEANS	Feb-90	0.53%	8.01%
WHEAT	May-03	5.93%	26.43%
<i>Equity index futures</i>			
EURO STOXX 50	Feb-99	3.52%	18.05%
FTSE 100	Feb-90	5.03%	13.76%
NIKKEI 225	Feb-90	3.14%	20.62%
S&P 500	Oct-97	8.24%	15.44%
<i>Bond futures</i>			
EURO-BUND	Dec-90	1.38%	5.85%
JAPAN 10-YEAR BOND	Feb-90	0.93%	3.86%
US 10-YEAR NOTE	Feb-90	0.59%	6.21%
<i>Currency futures</i>			
AUD/USD	Feb-90	0.36%	11.02%
EUR/USD	Jun-98	0.65%	9.27%
GBP/USD	Feb-90	-0.14%	8.97%
JPY/USD	Feb-90	0.37%	10.59%

Notes: The table reports key statistics for the futures in the sample.

4. Methodology

This chapter outlines this dissertation’s methodology. A key distinction is made throughout. Cross-asset risk normalisation scales each contract position by its own ex ante volatility so that no single market dominates portfolio risk, whereas volatility management adjusts overall strategy exposure as a function of the strategy’s own recent realised variance.

Monthly and daily returns are calculated as simple percentage changes in settlement prices. For contract i at month t , the monthly return is defined as:

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

where $P_{i,t}$ denotes the settlement price after each month. Daily returns are defined using daily settlement prices. To limit the influence of anomalies associated with contract roll mechanics, any return observation with $|r| \geq 0.8$ is set to missing. This filter is applied uniformly across all contracts and time periods.

The panel is unbalanced, because not every contract is available across the full sample. Portfolio returns are then calculated by using equal-weight averages across the contracts that have no missing observations, allowing the cross-sectional composition of the portfolio to vary over time without adding in artificial returns. Monthly returns are used for signal formation and portfolio evaluation, whilst daily returns are used solely for volatility estimation.

The time-series momentum signal follows Moskowitz, Ooi and Pedersen (2012). The position direction for each contract is determined by the sign of its own past 12-month return, such that assets with positive past performance are assigned long positions and those with negative past performance are assigned short positions. The signal is the sign of the previous 12 month return.

Positions are implemented with a one-month lag, so that the signal formed using information available through month $t - 1$ is applied to returns in month t . This ensures that the strategy remains free from look-ahead bias. The baseline TSMOM portfolio follows the volatility-scaling approach of Moskowitz, Ooi and Pedersen (2012), in which each contract’s position is scaled inversely by its own ex ante volatility estimate $\sigma_{i,t}$. This prevents more volatile contracts from dominating portfolio risk. Following Moskowitz, Ooi and Pedersen (2012), the constant volatility target is set at $\kappa = 0.40$. This is standard in the time-series momentum literature, and it is also applied across all assets, so that it can enter the strategy as a common scaling factor for each position. This means that κ only changes the level of returns and volatility within the portfolio, without affecting risk measures like the Sharpe ratio. The contract-level return contribution is:

$$\tilde{r}_{i,t}^{TSMOM} = \frac{\kappa}{\hat{\sigma}_{i,t-1}} \cdot \text{sign}(r_{i,t-12,t-1}) \cdot r_{i,t}$$

Both the signal and the volatility estimate are lagged by one period.

The portfolio return is then calculated as an equally weighted average of the contract level returns across all of the contracts. Because the number of available contracts varies over time, the portfolio allocates equal weight to each contract available in a given month rather than to a fixed cross-section, which is appropriate given the dissertation's emphasis on broad cross-asset evidence.

Ex ante volatility is estimated from daily returns using an exponentially weighted moving average, capturing persistence in volatility whilst relying entirely on past data. Following Moskowitz, Ooi and Pedersen (2012), the annualised variance estimate is given by:

$$\hat{\sigma}_{i,t}^2 = 261 \sum_{k=0}^{\infty} (1 - \delta) \delta^k (r_{i,t-k} - \bar{r}_i)^2$$

where $\delta = \frac{60}{61} = 0.984$ is the decay parameter, $r_{i,t}$ is the daily return, $\bar{r}_i$ is the exponentially weighted mean return, and the factor 261 annualises the variance using the assumed number of trading days per year. The month-end volatility estimate is taken as the final daily value within each month and is then lagged by one period when used in portfolio construction, again to eliminate look-ahead bias.

Portfolio-level volatility management follows Moreira and Muir (2017), who show that scaling strategy exposure inversely with lagged realised variance can improve risk-adjusted performance without requiring forecasts of expected returns. The intuition is that, if volatility is persistent over time, reducing exposure after highly volatile periods and increasing exposure during calmer conditions may generate a more stable payoff profile than maintaining constant exposure throughout.

The realised variance of the underlying momentum strategy is estimated each month using the squared daily returns of the signal-only equal-weight momentum portfolio within that month, averaged across trading days. The volatility-managed return then scales the monthly base return by the inverse of this lagged realised variance. Following Moreira and Muir (2017), the managed portfolio return is defined as:

$$f_t^\sigma = \frac{c}{\sigma_{t-1}^2} f_t$$

where f_t is the monthly return of the base momentum strategy, σ_{t-1}^2 is the lagged realised variance of that strategy, and c is a scaling constant defined as the ratio of the standard deviation of the base series to the standard deviation of the raw scaled series, computed over the common sample. This rescaling makes the managed strategy comparable to the time-series momentum strategy in terms of volatility, so any differences observed in performance are more likely to reflect actual differences in strategy efficiency rather than mechanical differences in average risk exposure.

It should be noted, however, that the volatility-managed series is constructed from a signal-only equal-weight momentum base rather than from the TSMOM portfolio defined earlier. The managed strategy should therefore be interpreted as a Moreira and Muir (2017) style managed variance momentum strategy, rather than as the baseline TSMOM portfolio with a pure volatility-management overlay. Differences between the two strategies thus reflect both portfolio-level variance timing and differences in the underlying base portfolio construction.

To see if the strategies produce any significant alpha, monthly excess returns are regressed on a set of standard risk factors. Excess returns are calculated against the 1-month Treasury bill rate, which acts as the risk-free rate in this paper, this is consistent with standard asset-pricing practice in Fama and French (1993). The core model estimated is:

$$r_t^S - r_t^f = \alpha^S + \beta_1 \text{MKT}_t + \beta_2 \text{BOND}_t + \beta_3 \text{GSCI}_t + \beta_4 \text{SMB}_t + \beta_5 \text{HML}_t + \beta_6 \text{UMD}_t + \varepsilon_t$$

where α^S denotes the regression model's alpha, which is interpreted as the abnormal performance, and the betas are the corresponding factor loadings. MKT, SMB, and HML represent asset-pricing factors commonly used in empirical finance papers. UMD is the Carhart (1997) momentum factor, and BOND and GSCI control for commodity and bond exposure. Full definitions provided in Table 1. The regression is estimated by ordinary least squares with Newey-West HAC standard errors using 12 monthly lags, which is appropriate given the potential for both serial correlation and conditional heteroskedasticity in strategy returns. Alpha estimates are reported at the monthly frequency. When annualised figures are depicted the approximation $12\alpha^S$ is used.

As a final step, the paper assesses whether the differences in the Sharpe ratios are important statistically. A circular block bootstrap test in the spirit of Ledoit and Wolf (2008) is used, and it is performed using monthly returns and resampling blocks of length $\sqrt{T}$, following Politis and Romano (1994).

5. Results

5.1 Gross Performance

Table 3 reports annualised summary statistics for the passive benchmark, the time-series momentum strategy (TSMOM), and the volatility-managed time-series momentum strategy (VM-TSMOM). TSMOM delivers an annualised mean return of 7.8%, compared with 4.7% for the passive benchmark. This generally aligns with the findings from Moskowitz, Ooi and Pedersen (2012), and Figure 1 reinforces it.

TABLE 3 – GROSS PERFORMANCE PASSIVE, TSMOM, and VM-TSMOM

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
Passive benchmark	4.7%	8.4%	0.554	-30.2%
TSMOM	7.8%	14.3%	0.544	-35.4%
VM-TSMOM	3.6%	6.7%	0.527	-19.5%

Notes: The table reports the gross performance statistics for the passive benchmark.

This return advantage, however, comes with a substantial increase in risk. The annualised volatility of TSMOM is 14.3%, compared with 8.4% for the passive benchmark, and its maximum drawdown reaches -35.4%, exceeding the -30.2% recorded by the benchmark. The cumulative return path of TSMOM is visibly more volatile, with several episodes of sharp decline that are consistent with the reversals discussed by Daniel and Moskowitz (2016). As a result, the Sharpe ratio of TSMOM is 0.544, only marginally below the 0.554 of the passive benchmark. Time-series momentum therefore enhances absolute returns without meaningfully improving the risk–return trade-off, since the additional return is largely proportional to the additional risk taken on.

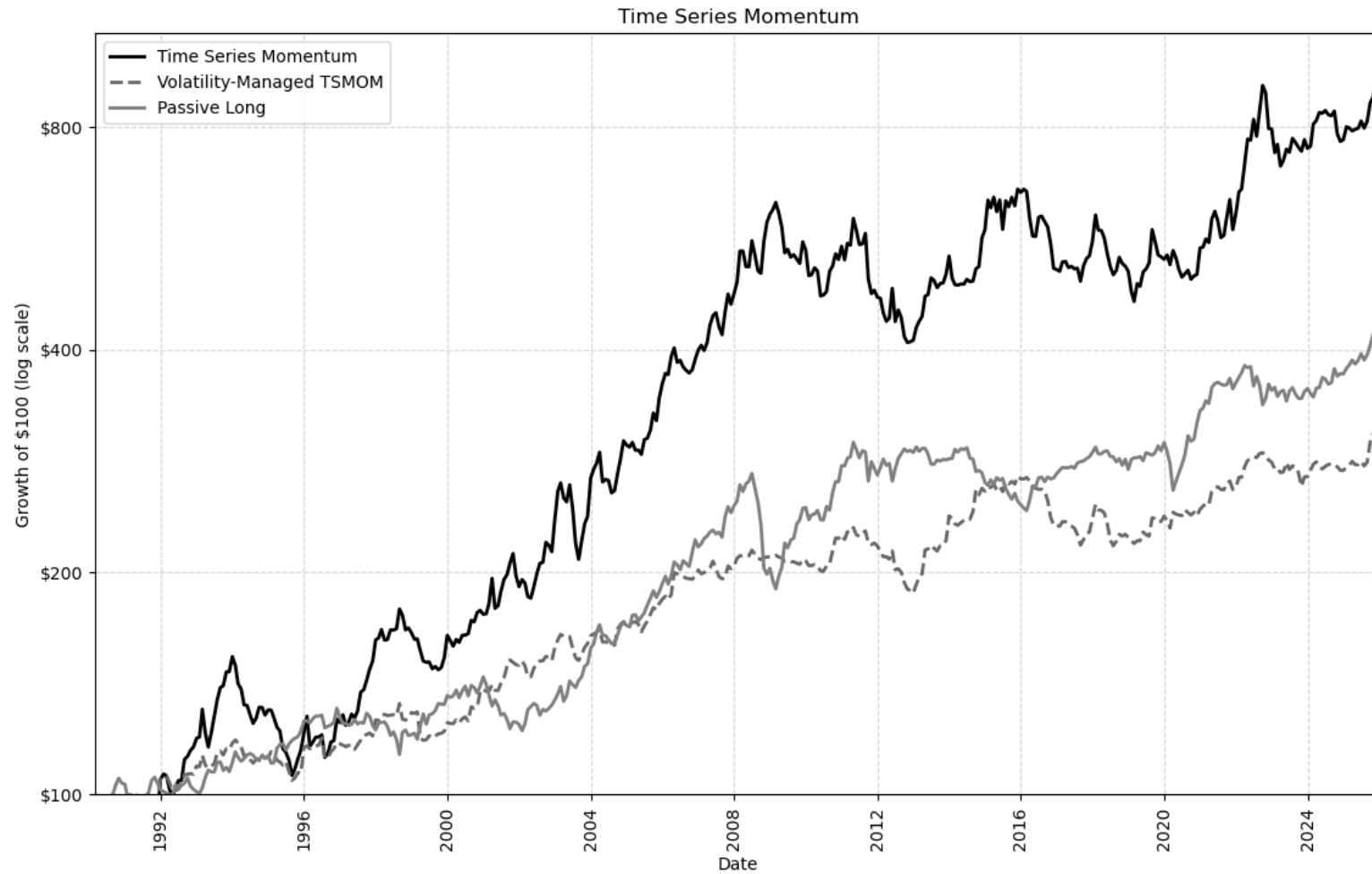
VM-TSMOM produces a markedly different return profile. It delivers an average return of 3.6%, around half the returns generated by TSMOM and below even the passive strategy, but its volatility drops to 6.7% and maximum drawdown to -19.5%. The cumulative growth of \$100 reaches around \$319 by the end of the sample, well below the other two strategies. Despite the lower return, the Sharpe ratio is 0.527, broadly in line with both alternatives. The managed strategy therefore trades lower returns for lower risk, leaving the efficiency of the return stream broadly unchanged in gross terms.

The behaviour of the VM-TSMOM scaling weight is informative in this respect. The median monthly weight applied to the base momentum portfolio is approximately 0.88, whilst the weight ranges from near zero in the most volatile months to over five in the calmest. The fact that the median weight sits below unity indicates that the strategy spends most of the sample at an effective exposure below that of the unscaled portfolio, which is consistent with its lower average return.

Before interpreting these comparisons, one methodological caveat should be made clear. As described in Section 4, the VM-TSMOM portfolio is constructed by applying variance scaling to an equally weighted momentum base that uses the sign of past returns to determine the trading position it takes. By contrast, the TSMOM portfolio applies asset-level volatility scaling across its cross-section as its own form of risk management. Therefore, both strategies are different in how variance is managed, but also in how the base portfolio is constructed. That said, the analysis does not isolate the effect of volatility management by itself, and as argued in Section 6, that works against the main finding. If variance timing does not provide a meaningful improvement to momentum even when the base portfolio is not already shaped by cross-asset volatility scaling, then the argument that overlaying volatility management does not materially improve strategy efficiency is more persuasive.

The gross-performance results show that time-series momentum continues to deliver meaningful return over a passive strategy benchmark, but that this return premium is linked to elevated volatility and drawdown risk. Volatility management substantially reduces risk, yet does so at the cost of a much lower average return, leaving the Sharpe ratios of all three strategies broadly comparable. Whether this reflects a genuine improvement in strategy efficiency, rather than simply lower risk exposure, is examined in the robustness checks below.

FIGURE 1 – CUMULATIVE GROWTH OF \$100 INVESTED IN, FOR THE PASSIVE, TSMOM, and VM-TSMOM STRATEGIES



Notes: The figure plots the cumulative value of \$100 invested in each strategy over the sample period using gross monthly returns.

5.2 Regression Results

Table 4 reports the factor regression results for both strategies. For TSMOM, the estimated alpha coefficient is 0.0024. However, the estimate is not statistically significant ($p = 0.344$), and the strategy therefore does not deliver a reliably positive return within this framework. Among the regressors, the only significant coefficient is UMD, which has a loading of 0.297 ($p < 0.001$). Although time-series momentum and cross-sectional momentum are conceptually distinct strategies, the positive and highly significant UMD loading shows that TSMOM returns move together meaningfully with the broader momentum premium. The remaining factor loadings are individually small and statistically insignificant, consistent with the view that a diversified trend-following strategy holding both long and short positions across asset classes has limited net directional exposure to any single one. The R-squared of 0.118 indicates that the included factors explain a modest share of the variation in TSMOM returns, with most of the explanatory power attributable to UMD.

For VM-TSMOM, the results are qualitatively similar but weaker in magnitude. The estimated alpha coefficient is slightly negative at -0.0004 and again insignificant ($p = 0.736$), providing no evidence of abnormal performance. The UMD coefficient remains the only significant loading, but at 0.117 it accounts for around 40% of the corresponding TSMOM estimate, indicating that volatility management softens the strategy's exposure to the momentum premium rather than amplifying it. The remaining factors are insignificant, and the R-squared falls to 0.078.

Altogether, the regression results warrant closer interpretation. The small and mostly insignificant coefficients for MKT, BOND, SMB, and HML are in line with how the strategies are constructed. A diversified long-short momentum strategy is expected to have limited net directional exposure to any one asset class as a whole, because the long and short positions partially offset each other at the portfolio level. The relatively low R^2 supports this intuition, as it suggests that a standard linear regression model explains only a fractional share of the variation in momentum returns, with most of that explanatory power coming from the coefficient *UMD*. This is consistent with the longstanding view in Fung and Hsieh (2001) which is that the payoffs of trend-following strategies resemble lookback straddles, which implies that a standard linear regression model is unlikely to capture the structure of momentum returns effectively. Hence, an insignificant alpha within this model specification should be treated with caution. Therefore, no statistically significant alpha is observed in this sample, but that does not mean that the strategies hold no economic value. Together with the much smaller *UMD* loading under volatility management, the regression results support the view that variance timing dampens momentum without producing offsetting efficiency gains.

TABLE 4 – FACTOR REGRESSIONS FOR TSMOM AND VM-TSMOM

	(1)	(2)
<i>VARIABLES</i>	<i>TSMOM</i>	<i>VM-TSMOM</i>
Alpha	0.0024 (0.0025)	-0.0004 (0.0012)
MKT	-0.0019 (0.0776)	0.0295 (0.0267)
BOND	0.1334 (0.3436)	0.1176 (0.0970)
GSCI	0.0073 (0.0526)	0.0066 (0.0260)
SMB	-0.0926 (0.0696)	-0.0295 (0.0309)
HML	0.0383 (0.0596)	0.0341 (0.0308)
UMD	0.2971*** (0.0559)	0.1166*** (0.0228)
Observations	419	418
R-squared	0.1182	0.0776

Notes: Newey–West HAC standard errors in parentheses (12 lags). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 Robustness checks

5.3.1 Return-matched comparison

The gross performance results in Section 5.1 show that VM-TSMOM has a lower average return alongside lower volatility and a reduced maximum drawdown. However, comparing how risk is managed between two strategies that operate at different average levels of risk exposure is misleading since a strategy that takes on less risk will automatically appear more stable without being any more efficient. A more informative comparison is therefore one that is performed solely on the VM-TSMOM return series. This involves evaluating the strategy at its natural exposure against the same strategy scaled by a constant in order to match the average return of TSMOM. This analysis will show the effect of running variance timing at a higher level of exposure, and this would be independent of how the underlying base portfolios are constructed.

The constant needed to scale to TSMOM is 2.183, which implies that VM-TSMOM in its unscaled form produces around half of the monthly return of the TSMOM strategy. Once it is matched to TSMOM, the apparent efficiency improvement fades. The mean volatility over time of return-matched VM-TSMOM goes up to 14.7%, which is only slightly above TSMOM's 14.3%, and its Sharpe ratio falls to 0.527 lower than 0.544 for TSMOM. Most dramatically, the maximum drawdown worsens to -38.9%, also worse than the -35.4% of TSMOM.

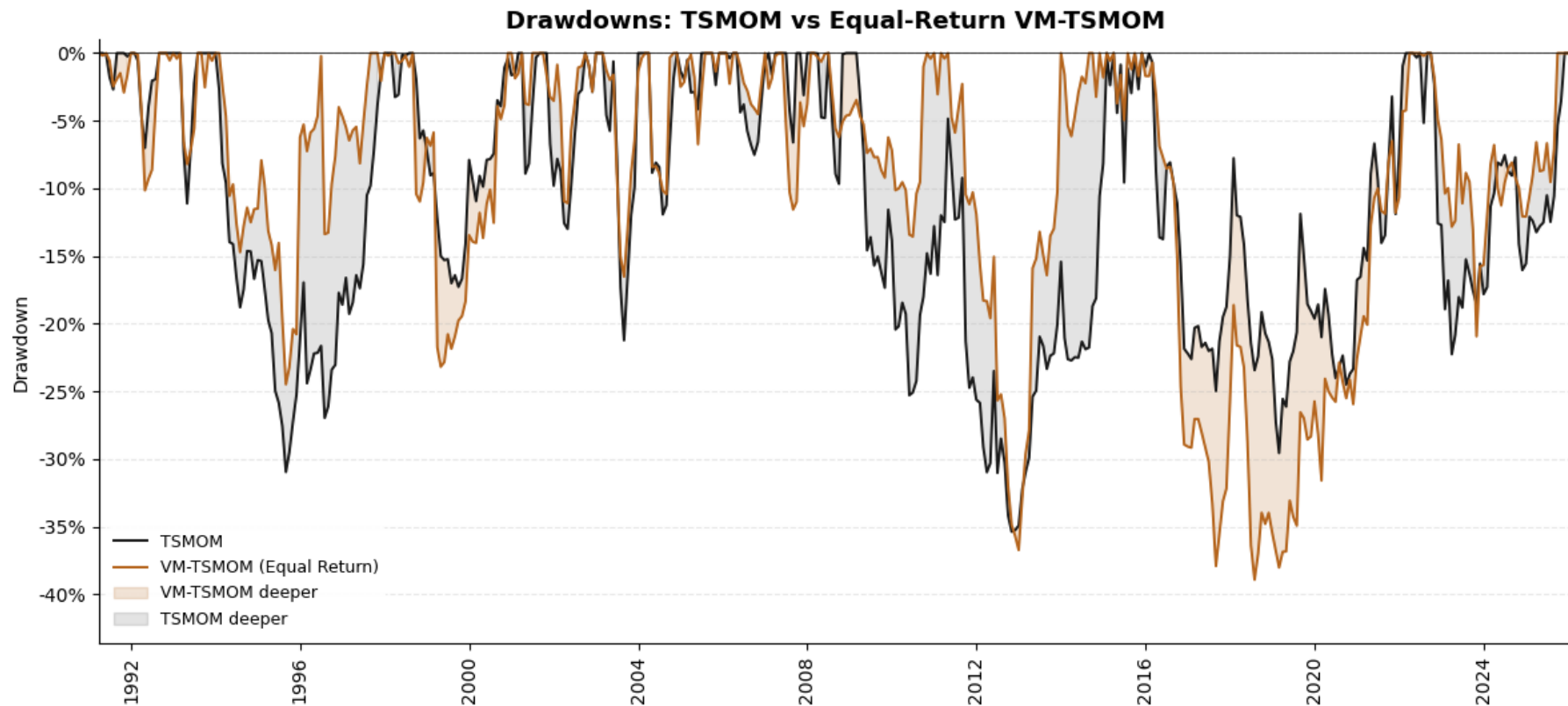
TABLE 5 – PERFORMANCE COMPARISON TSMOM & RETURN-MATCHED VM-TSMOM

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
TSMOM	7.8%	14.3%	0.544	-35.4%
Return-matched VM-TSMOM	7.8%	14.7%	0.527	-38.9%

Notes: Return-matched VM-TSMOM is constructed by multiplying the original VM-TSMOM return series by a constant so that its sample average return equals that of TSMOM.

Figure 2 provides an informative visualisation and shows that after the rescaling, the VM-TSMOM portfolio compared to TSMOM dips further during sharp drawdown episodes. The more stable return path from the gross comparison is therefore the result of operating the strategy at lower average levels of risk, and not due to any genuine improvement in how risk is managed. As soon as the strategies are required to produce the same returns, variance timing leads to a worsening in drawdowns and provides no competitive efficiency advantage. This provides evidence that volatility management here is more a form of taking on less risk than a way to improve risk-adjusted returns.

FIGURE 2 – DRAWDOWNS OF TSMOM AND RETURN-MATCHED VM-TSMOM



Notes: The figure plots the drawdown paths of TSMOM and the return-matched VM-TSMOM strategy. VM-TSMOM is rescaled so that its sample average return matches that of TSMOM, allowing downside risk to be compared at a common return level.

5.3.2 Transaction costs

The analysis so far has been conducted on gross returns and does not account for the practical costs of implementing either strategy. Both TSMOM and VM-TSMOM require periodic rebalancing as signals and volatility estimates change, and the managed strategy in particular involves additional adjustments in exposure driven by movements in lagged realised variance. To assess the sensitivity of the results to these frictions, one-way monthly portfolio turnover is computed for each strategy as half the sum of the absolute changes in portfolio weights from one month to the next. Net returns are then calculated by deducting transaction costs based on turnover, using 5 and 10 basis points as the cost assumptions.

The turnover estimates in Table 6 confirm that the volatility-managed strategy is substantially more trading-intensive. Average monthly one-way turnover for TSMOM is 0.614, compared with 1.715 for VM-TSMOM, nearly three times as high. The variance-timing mechanism adjusts the level of portfolio exposure each month in response to changes in lagged realised variance, generating additional weight changes on top of those driven by the monthly signal updates that affect both strategies equally.

TABLE 6 – TURNOVER of TSMOM and VM-TSMOM

Strategy	Average monthly turnover	Median monthly turnover
TSMOM	0.614	0.522
VM-TSMOM	1.715	1.235

Notes: Monthly turnover is calculated as half the sum of absolute changes in portfolio weights from one month to the next.

The impact of transaction costs is markedly asymmetric. For TSMOM, the annualised return falls modestly from 7.8% gross to 7.4% under 5 basis points and 7.0% under 10 basis points, with the Sharpe ratio declining from 0.544 to 0.518 and 0.492 respectively. The maximum drawdown worsens only slightly. The baseline strategy therefore remains economically viable under reasonable cost assumptions for liquid futures markets. VM-TSMOM, by contrast, deteriorates far more severely. Its annualised return falls from 3.6% gross to 2.5% under 5 basis points and to just 1.5% under 10 basis points. The Sharpe ratio drops from 0.527 to 0.375 and 0.222, whilst the maximum drawdown worsens from -19.5% to -25.3% and -31.2%.

TABLE 7 – PERFORMANCE OF TSMOM AND VM-TSMOM AFTER COSTS

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
TSMOM (5 bps)	7.4%	14.3%	0.518	-36.2%
TSMOM (10 bps)	7.0%	14.2%	0.492	-36.9%
VM-TSMOM (5 bps)	2.5%	6.7%	0.375	-25.3%
VM-TSMOM (10 bps)	1.5%	6.7%	0.222	-31.2%

Notes: Net returns are calculated by deducting costs of 5 and 10 basis points using monthly turnover.

These results considerably weaken the case for the volatility-managed strategy. At 10 basis points, its Sharpe ratio is less than half that of the net TSMOM portfolio, and its maximum drawdown approaches the level observed for TSMOM on a gross basis. The smoother gross-return profile documented in Section 5.1 is therefore not only conditional on accepting lower average returns, as the return-matched comparison shows, but is also substantially more expensive. Given that the managed strategy already fails to outperform TSMOM on a return-matched basis, the further decline from the added transaction costs reinforces the conclusion that volatility management does not improve the investability of time-series momentum in this sample period.

It should be acknowledged that this cost model is a stylised approximation. In practice, futures transaction costs vary across assets and over time. The 5 and 10 basis point assumptions are intended to represent a plausible range for liquid futures markets rather than a precise estimate of realised costs. Nonetheless, the broader conclusion remains clear: because VM-TSMOM turnover is nearly three times that of TSMOM, even modest cost assumptions are sufficient to erode its already limited advantage.

5.3.3 Subperiod analysis: pre-2009 vs post-2009

A final concern is whether the results are regime-dependent and therefore conceal meaningful changes in strategy behaviour over time. The post global financial crisis (GFC) period has been marked by unconventional monetary policy, compressed yields, and an environment in which trend-following has been widely reported to underperform compared to its earlier history. To examine this, the sample is split into a pre-2009 period and a post-2009 period, with both summary statistics and the factor regressions re-estimated within each subperiod.

TABLE 8 – SUBPERIOD PERFORMANCE

Strategy	Annual return	Volatility	Sharpe ratio	Maximum drawdown
Pre-2009 period				
Passive	4.3%	8.3%	0.519	-26.9%
TSMOM	11.2%	13.7%	0.821	-31.0%
VM-TSMOM	4.4%	6.6%	0.664	-11.8%
Post-2009 period				
Passive	5.0%	8.5%	0.589	-19.2%
TSMOM	4.1%	14.8%	0.278	-35.4%
VM-TSMOM	2.7%	6.9%	0.390	-19.5%

Notes: The sample is split into pre-2009 and post-2009 subperiods to assess whether performance differs across market environments. Reported returns and volatilities are annualised. Maximum drawdown is reported over each respective subperiod.

The two subperiods differ substantially. In the pre-2009 period, TSMOM delivered an annualised return of 11.2% with a Sharpe ratio of 0.821, comfortably exceeding the passive benchmark's 4.3% return and 0.519 Sharpe ratio. VM-TSMOM also performed well in relative terms, with a return of 4.4% and a Sharpe ratio of 0.664, supported by a modest maximum drawdown of -11.8%. In the post-2009 period, however, the picture changes substantially. The return on TSMOM collapses to 4.1%, and its Sharpe ratio falls to 0.278, even as its volatility rises slightly. The maximum drawdown worsens to -35.4%, accounting for almost all of the full-sample drawdown figure reported earlier. VM-TSMOM also

weakens, with a return of 2.7% and a Sharpe ratio of 0.390. By contrast, the passive benchmark improves modestly, with a Sharpe ratio of 0.589. The relative attractiveness of trend-following therefore deteriorates sharply in the post-2009 environment, whilst a simple long-only portfolio holds up well.

The subperiod regressions, reported in Table 9, reinforce this picture but add important nuance. The TSMOM alpha is essentially zero in the pre-2009 period and 0.0029 in the post-2009 period, but neither estimate is statistically significant. Consequently, there is no evidence of abnormal returns from TSMOM in either subperiod. The UMD loading is positive and significant in both subperiods, rising from 0.229 pre-2009 to 0.343 post-2009, which suggests that TSMOM has become more closely tied to the broader cross-sectional momentum premium in the more recent sample. The pre-2009 BOND loading of 0.800 is also significant, reflecting the importance of long bond positions during the disinflationary period that makes up much of the early sample, but this is no longer significant post-2009. Other factor coefficients remain small in both subperiods.

For VM-TSMOM, the only material change is in the pre-2009 alpha, which has a coefficient of -0.0025 with a small standard error and is statistically significant. This negative point estimate should be interpreted cautiously: it indicates that, in the earlier subperiod, the variance-managed strategy delivered returns slightly below what its factor exposures alone would imply, rather than evidence of any economic underperformance once the lower exposure of the strategy is taken into account. The post-2009 alpha is again practically zero. The UMD loading remains positive and significant in both subperiods, but at roughly 0.11 it is consistently smaller than the TSMOM loading, in line with the full-sample finding that volatility management dampens rather than amplifies exposure to the momentum premium.

The broader conclusion from the subperiod analysis is that the central findings of this dissertation are not driven by any single regime. In neither subperiod does TSMOM produce a statistically significant alpha, and in neither subperiod does volatility management deliver an abnormal return that would justify it on efficiency grounds. The principal change across subperiods is the marked weakening of trend-following performance in the post-2009 environment, which affects both strategies and is most visible in the collapse of the TSMOM Sharpe ratio. The robustness of the qualitative conclusions across the two halves of the sample lends additional confidence to the interpretation developed in the earlier sections: time-series momentum remains a meaningful source of return relative to a passive benchmark on average, but it does not generate reliable risk-adjusted abnormal performance, and volatility management does not improve its investability.

TABLE 9 – SUBPERIOD FACTOR REGRESSIONS, PRE-2009 vs POST-2009

	(1)	(2)	(3)	(4)
<i>VARIABLES</i>	<i>TSMOM Pre-2009</i>	<i>TSMOM Post-2009</i>	<i>VM-TSMOM Pre-2009</i>	<i>VM-TSMOM Post-2009</i>
Alpha	-0.0000 (0.0027)	0.0029 (0.0032)	-0.0025** (0.0010)	0.0008 (0.0018)
MKT	-0.0431 (0.1122)	0.0947 (0.1022)	0.0198 (0.0331)	0.0705 (0.0437)
BOND	0.7995** (0.3446)	-0.4937 (0.4244)	0.2712** (0.1171)	-0.0474 (0.1370)
GSCI	0.0063 (0.0553)	-0.0353 (0.0883)	0.0103 (0.0381)	-0.0096 (0.0347)
SMB	-0.0726 (0.0940)	-0.0879 (0.1225)	0.0091 (0.0392)	-0.0626 (0.0612)
HML	-0.0223 (0.1109)	0.0440 (0.0745)	0.0807 (0.0534)	-0.0015 (0.0357)
UMD	0.2288*** (0.0695)	0.3434*** (0.0830)	0.1092*** (0.0296)	0.1151*** (0.0395)
Observations	214	204	214	204
R-squared	0.1500	0.1449	0.1190	0.0699

Notes: Newey–West HAC standard errors in parentheses (12 lags). *** p < 0.01, ** p < 0.05, * p < 0.1

5.3.4 Testing Sharpe ratio differences

Whilst the analysis compares the Sharpe ratios between the different strategies in this dissertation, it does not assess whether these differences are of statistical significance. Table 10 directly addresses this. It reports a circular block bootstrap test of equal Sharpe ratios relative to TSMOM. The difference between the gross Sharpe ratio of TSMOM and VM-TSMOM is small and insignificant, which is consistent with the earlier results of volatility management not improving risk-adjusted performance once differences in strategy exposure are considered. This also holds after a 5 basis point transaction cost assumption. Under 10 basis points, the Sharpe ratio difference between the two strategies becomes statistically significant ($p = 0.015$), indicating that the sharp decline in VM-TSMOM performance is economically and statistically significant once transaction costs are introduced, however in a negative sense.

TABLE 10 – BOOTSTRAP TESTS OF SHARPE RATIO DIFFERENCES RELATIVE TO TSMOM

Comparison	SR1	SR2	Difference	p-value
TSMOM vs VM-TSMOM	0.544	0.527	0.017	0.876
TSMOM vs return-matched VM-TSMOM	0.544	0.527	0.017	0.876
TSMOM (5 bps) vs VM-TSMOM (5 bps)	0.518	0.375	0.143	0.189
TSMOM (10 bps) vs VM-TSMOM (10 bps)	0.492	0.222	0.270	0.015*

Notes: p-values obtained from a circular block bootstrap test in the spirit of Ledoit and Wolf (2008) is used, and it is performed using monthly returns and resampling blocks of length $\sqrt{T}$, following Politis and Romano (1994). The SR is the Sharpe ratio.

An additional robustness check uses alternative lookback periods of 1, 3, 6, 9 and 12 months. This is done for both TSMOM and VM-TSMOM and is reported in the Appendix. The conclusions from the main analysis remain unchanged after the test across all specifications. TSMOM delivers positive average returns across all of the time horizons. VM-TSMOM does not improve risk-adjusted performance relative to TSMOM across any lookback. The 12-month horizon used throughout the main results of the dissertation delivers the strongest performance for TSMOM, this is consistent with the results in Moskowitz, Ooi and Pedersen (2012) and with Novy-Marx's (2012) argument that momentum profitability is concentrated in medium return horizons.

6. Discussion

6.1 Interpretation of the Main Findings

Why variance timing fails to improve strategy efficiency in this particular setting is a natural question, when it appears to do so for equity factor portfolios. It is likely that it is because of how the risk is already managed in the base strategy. The TSMOM strategy uses inverse-volatility scaling at the individual asset level, preventing a particular asset from dominating the entire portfolio and reducing the available residual variance a portfolio-level overlay would be able to exploit. The equity factor portfolios that are studied by Moreira and Muir (2017) carry no comparable cross-sectional risk normalisation, leaving more space for a variance-scaling overlay to work with. The null result documented here is not inconsistent with their findings, it suggests that variance timing works as a substitute to, rather than an addition to, cross-asset risk normalisation.

The regression results point to the same conclusion but from a different angle. Volatility management does not lead to positive alphas and it reduces the strategy's exposure to cross-sectional momentum. Interpreted through Fung and Hsieh (2001), this suggests that the overlay reduces the option-like payoff that makes trend-following unique, rather than refining it or making it more stable. If a strategy earns returns through a non-linear payoff structure, an overlay that is based upon linear variance signals is unlikely to contribute to additional value.

The results from the subperiod analysis add to the discussion. After 2009 both strategies experienced similar weakening effects, the deterioration can be interpreted as a result of the underlying momentum signal. If variance timing could correct regime change, then you would expect that the gap between TSMOM and VM-TSMOM relative performance would widen here, but it does not.

6.2 Comparison with the Literature

The findings neither contradict nor confirm the core literature. Instead, they relate to it in a more nuanced way. The existing presence of return continuation in this dissertation is consistent with Moskowitz, Ooi and Pedersen (2012), and with the extended long-run evidence that is reported by Hurst, Ooi and Pedersen (2017). The key difference from Moskowitz, Ooi and Pedersen (2012) is in the strength of the return premium rather than the existence of it. In their original study, time-series momentum delivers a Sharpe ratio above 1.0 and large, statistically significant alphas. In contrast, in this dissertation the Sharpe ratios lie much closer to the passive benchmark and it fails to produce statistically significant alphas. This is also consistent with the critique of Kim, Tse and Wald (2016), who argue that a significant share of the strategy's profitability stems from the volatility-scaling used rather than time-series momentum's predictability of returns.

This considerably weaker result is likely due to numerous factors, including a narrower number of assets used, using a later sample that captures the weaker trend-following performance following the GFC period, and choices in how the strategy was implemented such as the type of data used and equal weight aggregations.

Comparing the results to Moreira and Muir (2017) is particularly intriguing. Their main finding is that scaling exposure inversely with lagged realised variance can improve the performance of a strategy's Sharpe ratio and generate a statistically significant alpha, but this pattern is not evident within this dissertation. The apparent benefits in this setting, appear to come mainly from deploying the strategy at a lower average level of exposure rather than exploiting volatility predictability in a way that improves strategy efficiency. A reason for the difference could be because of what asset classes the paper studies. Moreira and Muir look at equity factor portfolios, whereas the strategies examined here are futures from various asset classes in which the cross-asset risk normalisation does much of the risk management at the base level, leaving less unmanaged variance for a portfolio level overlay to improve upon.

Daniel and Moskowitz (2016) complement Moreira and Muir (2017). Their emphasis on the state-dependent crash risk of momentum strategies is consistent with the deeper drawdowns observed for TSMOM in this sample. The reduction in the *UMD* coefficient under volatility management is related to the risk exposure reduction they mention. Their broader view, however, is that crash risk is only partly captured by realised variance alone, which is a useful warning against expecting simple variance scaling to deliver crash protection. The fact that drawdowns do not improve once returns are matched is consistent with that view.

As a final nod to the literature, the weaker momentum performance that is observed after 2009 could be attributed to the post-publication decay argument from McLean and Pontiff (2016). However, their analysis examines anomalies using cross-sectional equity data, which heavily differs from this dissertation's use of cross-asset futures for analysing time-series momentum, so it would go too far to treat their findings as directly applicable in this scenario. Even then, the comparison is useful because it raises thoughts around why testing prominent strategies in a contemporary sample is necessary and relevant.

6.3 Limitations

Several limitations need to be considered when interpreting the results of this dissertation. The most direct is how the specific form of volatility management in this paper is tested. This was elaborated on in Sections 1 and 4, where a Moreira and Muir (2017) style variance-managed momentum strategy is evaluated, rather than an overlay on the cross-asset TSMOM portfolio. As argued above, this works against the main finding, but a cleaner test with a fixed base portfolio with a varying overlay would allow for a better evaluation of variance timing.

The way the data is constructed presents a possible limitation. The analysis is reliant on Bloomberg continuous futures prices and because of this the return data is dependent on how Bloomberg constructs those series, for example how contracts are rolled and adjusted. This

can have an impact on the value of the returns and the momentum signal itself. The filter that was applied removes monthly returns with an absolute value greater than 0.8 to limit the possible contract rolling anomalies. However, this could exclude genuine large moves that occur in volatile periods. The way transaction costs were calculated is similarly simplified. Using a basis point charge on turnover ignores several factors and whilst the conclusion that VM-TSMOM is more sensitive to costs is likely to remain, the size of that effect should be treated as an approximation and not exact.

The regression model also has potential limitations. The loading on UMD is formed from cross-sectional momentum on equities, therefore it captures a co-movement rather than an explanation of where the returns in the strategies are formed from. Generally, trend-following strategies are described as having payoffs that are option-like, which means that linear regressions may not fully reflect the underlying risk behind them. For this reason, the lack of significantly positive alpha in this dissertation should be interpreted as no significant alpha in this framework, rather than there being a lack of economic value. The regime split is a sensible way to examine differences in regimes, however it introduces a single break on what is likely a gradual change in economic conditions. Additionally, splitting the sample reduces the statistical power by having fewer observations, making it even harder to detect any remaining alpha even if it is present.

7. Conclusion

7.1 Main Findings

The aim of this dissertation was to examine time-series momentum's effectiveness as a trading strategy in a contemporary setting, and whether volatility management can be used to improve it.

Time series momentum does continue to deliver meaningful returns over a passive investment benchmark, however this does not come with clear improvements in risk-adjusted returns, and when factor regressions are run neither strategy generates a statistically significant alpha. The answer to the second question is more direct. When returns are matched between the two tested strategies the comparison shows that the supposed benefits of volatility management fade away once returns are made comparable, whilst the transaction cost analysis shows that any remaining advantages are fragile even under modest cost assumptions. The regime split gives more context without altering the broader conclusions, momentum weakens in the later period, but the main qualitative findings remain consistent across both halves of the sample.

These findings remain relevant alongside more recent literature that has begun to question the Moskowitz, Ooi and Pedersen (2012) time-series momentum results and the broader works on variance-management that is hypothesised as an improvement to the strategy. Huang, Li, Wang and Zhou (2020) directly challenge the robustness of the time-series momentum paper by Moskowitz, Ooi and Pedersen (2012), whilst Kim, Tse and Wald (2016) attribute much of the profitability of the strategy to the volatility-scaling specification, and Cederburg, O'Doherty, Wang and Yan (2020) showing how fragile variance-managed factor strategies are when they are taken out of sample. Therefore, the contribution of this dissertation is that it acts as an extension to the scepticism surrounding multi-asset momentum strategies, where the cross-asset risk normalisation already does a lot of the work a variance overlay on the underlying portfolio would otherwise be expected to do.

7.2 Implications

For an investor considering volatility management as a way to improve the performance of time-series momentum and make it a more investable strategy, the evidence suggests that using volatility management in this way mainly acts as a form of avoiding risk rather than a source of improving return efficiency.

Academically speaking, the results suggest that caution is needed when making attempts to extend the variance-management literature too broadly. The findings that seem to hold in equity factor settings do not necessarily translate to multi-asset future portfolios, where cross-asset risk normalisation is a good enough form of risk management.

7.3 Future Research

Two extensions arise most naturally from the limitations set out in Section 6.3. A cleaner test of volatility management could be implemented where the base portfolio remains fixed and varies only on the overlay, so the difference between the strategies is based on variance timing rather than how the portfolio is constructed. Another is to use a more detailed cost assumption that reflects differences in liquidity and the cost of rolling futures contracts, this would give a more convincing test of whether VM-TSMOM remains especially sensitive to costs under more realistic conditions.

8. References

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9. Appendix

Appendix A. Alternative Lookback Period Results

TSMOM Lookback Sensitivity

Strategy	Annual Return	Volatility	Sharpe Ratio	Max Drawdown
TSMOM (1m)	2.32%	14.00%	0.165	-67.33%
TSMOM (3m)	5.17%	14.23%	0.364	-46.51%
TSMOM (6m)	3.70%	14.47%	0.256	-45.26%
TSMOM (9m)	3.61%	13.39%	0.270	-39.18%
TSMOM (12m)	7.67%	14.25%	0.538	-35.37%

VM-TSMOM Lookback Sensitivity

Strategy	Annual Return	Volatility	Sharpe Ratio	Max Drawdown
VM-TSMOM (1m)	-0.31%	6.82%	-0.045	-50.26%
VM-TSMOM (3m)	1.03%	7.02%	0.147	-32.02%
VM-TSMOM (6m)	0.55%	7.04%	0.079	-31.99%
VM-TSMOM (9m)	0.65%	6.59%	0.099	-30.43%
VM-TSMOM (12m)	3.55%	6.74%	0.527	-19.47%

Appendix B. Data Clarification

The Bloomberg data used in this dissertation are generic first continuous futures series used for research and return construction. They are not single tradeable contracts as written, but rolling reference series linked to the underlying listed futures contracts.